

Bayes Rule:- describe how to update the Probability of a certain event after gaining new knowledge that affects the probability of the event.

$$P(A \cap B) = P(A|B)P(B)$$

$$P(B \cap A) = P(B|A)P(A)$$

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$$P(A|B)P(B) = P(B|A)P(A)$$

likelihood

$$\therefore P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Prior

Posterior $\leftarrow$

$P(B)$

marginal

This framework leads directly to:

- $\rightarrow$ Kalman filter
- $\rightarrow$ Extended Kalman filter
- $\rightarrow$ Unscented Kalman filter
- $\rightarrow$ Particle filter
- $\rightarrow$ Monte Carlo localization

Sensor Fusion:- it's about the development of a logic that receives multiple signals from different data sources, filters them and combine them in such a way the resulting signal is an improved version of all the ones received as input.

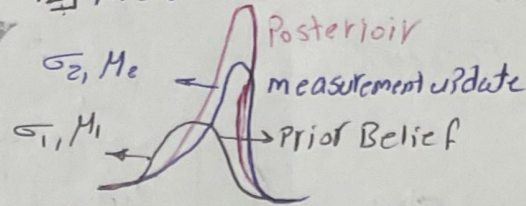
Kalman Filter

K.F.

Initial Estimate

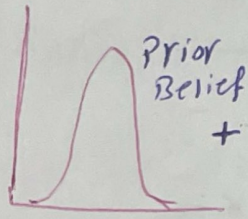


[2] measurement update

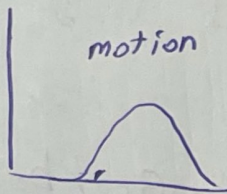


$$\sigma_3^2 = \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} \quad M_3 = \frac{\sigma_2^2 M_1 + \sigma_1^2 M_2}{\sigma_2^2 + \sigma_1^2}$$

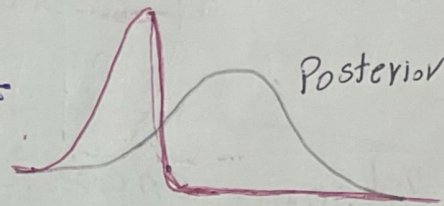
[3] state prediction



$$M = M_1 \quad \sigma^2 = \sigma_1^2$$



$$M = M_2 \quad \sigma^2 = \sigma_2^2$$

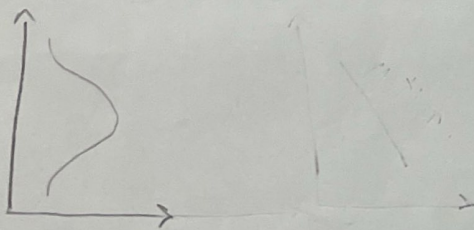


$$M_3 = M_1 + \mu$$

$$\sigma_3^2 = \sigma_1^2 + \sigma_2^2$$

Extended kalman filter

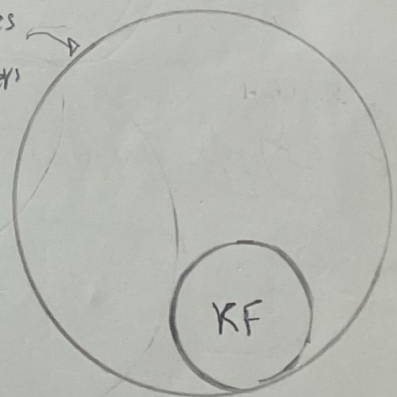
EKF



$$\frac{P(A|B)P(B|A)}{P(B)} \quad \text{Bayes Filter}$$

→ Linear System

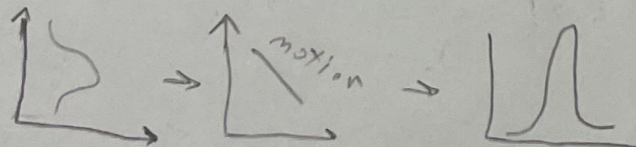
→ Gaussian Distributions



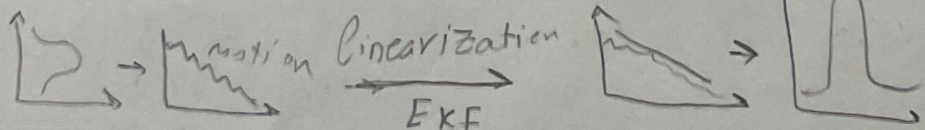
EKF is used to since it takes into account non-linearities in the motion model to understand how the EKF works

1st apply a non-linear function to Gaussian distribution

→ linear



→ non linear



Initial state → linearization → Measurement update loops → State Prediction

[15]